

Single-node mode: why the MIP bypass was wrong

OLD: HAND-FIX EVERYTHING

- 1x1 NoC bypassed the MIP solver
- hand-fixed node tile size (node_dims)
- ALSO hand-fixed DRAM order (dram_order)

1

HARDWARE ONLY FIXES CAPACITY

- node has a fixed per-dim capacity - real
- DRAM streaming ORDER is not fixed by hw
 - that's still a search problem

2

REAL MIP CONSTRAINT

- node_dim_capacity (node_capacity.py)
- bounds what fits at NodeLevel only
- solver stays free to search DRAM order

3

NET EFFECT

- single_node = pure topology flag
- no longer hijacks the solve
- schedule is always fully MIP-solved

4